

Stimulatory Effects of (+)-Epicatechin on Mitochondrial Biogenesis and Function in Skeletal Muscle of Aged Rats: Underlying Mechanisms

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Abstract

Mitochondrial dysfunction affects skeletal muscle (SkM) function and is critical in the etiology of age-related sarcopenia. The sirtuin 1-PGC1- α pathway is a key regulator of mitochondrial mass, structure, and function. However, pathway activity decreases with aging. Cacao flavanols show promise in their ability to activate mitochondrial pathways. We evaluated the capacity of the flavanol (+)-epicatechin (+Epi) to stimulate such a pathway and favorably impact mitochondrial and oxidative stress (OS)-associated endpoints in aged SkM. Using 23-month-old male Sprague-Dawley rats, an 8-week oral administration of +Epi (1 mg/kg/day) was implemented, and results were compared versus vehicle-treated controls. Assessments included the nicotinamide adenine dinucleotide (NAD)/sirtuin 1/Pgc1- α pathway, acetylated proteins levels, mitochondrial function and biogenesis, as well as OS-related endpoints in SkM. +Epi increased the NAD/NADH ratio, activation of sirtuin 1, the deacetylation of nuclear protein content, including that of PGC1- α . Also, +Epi induced increases of TFAM and NRF1 mRNA levels, deacetylation of mitochondrial complex I and V, increases in complex I activity, sirtuin 3, complexes I and V, mitofillin, and TFAM protein levels. SkM citrate synthase activity and ATP content increased with +Epi. OS markers in proteins and lipids were reduced, while buffering systems (superoxide dismutase 2 and catalase protein and activities) increased. In white blood cells, we documented serial reductions in mitochondrial DNA content and citrate synthase activity with aging, which were either fully or partially reversed with +Epi. Results demonstrate that +Epi treatment yields positive effects on mitochondrial biogenesis and function,

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